

Python CSV File Handling Practice

Menu Driven Programs

General Instructions

- Use only the built-in `csv` module.
- All data must be stored in **CSV files** (.csv).
- Programs must be **menu driven**.
- Handle files properly using `with` statement.

Program 1: Student Record Management System

File Name: `students.csv`

CSV Fields:

```
Roll No, Name, Course, Marks
```

Menu Options:

- 1. Add student record
- 2. View all student records
- 3. Exit

Task for Students:

- Create the CSV file if it does not exist.
- Write student data using `csv.writer`.
- Read and display all records using `csv.reader`.

Program 2: Employee Salary Register

File Name: `employees.csv`

CSV Fields:

```
Emp ID, Name, Department, Salary
```

Menu Options:

- 1. Add employee
- 2. Display all employees
- 3. Search employee by ID
- 4. Exit

Task for Students:

- Append new employee data.
- Search and display employee details based on ID.

Program 3: Product Inventory System

File Name: products.csv

CSV Fields:

```
Product ID, Product Name, Price, Quantity
```

Menu Options:

- 1. Add product
- 2. View inventory
- 3. Display products with quantity below a given value
- 4. Exit

Task for Students:

- Read numeric values properly from CSV.
- Apply conditions while reading CSV data.

Program 4: Attendance Management System

File Name: attendance.csv

CSV Fields:

```
Date, Roll No, Name, Status
```

Menu Options:

- 1. Mark attendance

- 2. View attendance by date
- 3. View full attendance record
- 4. Exit

Task for Students:

- Append daily attendance records.
- Filter records based on date.

Program 5: Library Management System (Edit & Delete)

File Name: library.csv

CSV Fields:

```
Book ID, Book Name, Author, Status
```

Menu Options:

- 1. Add book
- 2. View all books
- 3. Update book status (Issued / Available)
- 4. Delete book record
- 5. Exit

Important Instructions:

- Editing and deleting must be done by:
 - Reading all records into a list
 - Modifying the list
 - Writing the updated list back to the CSV file
- Do NOT edit the CSV file directly.

End of CSV Practice Problems